# ANDERSON JUNIOR COLLEGE

## 2017 JC 2 PRELIMINARY EXAMINATIONS

**CHEMISTRY** **9729/03**

Paper 3 Free Response

**15 September 2017**

**2 hours**

Candidates answer on separate paper.

Additional Materials: Answer Paper
Data Booklet

**READ THESE INSTRUCTIONS FIRST**

Write your name, PDG and register number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

**Section A**
Answer **all** questions.

**Section B**
Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

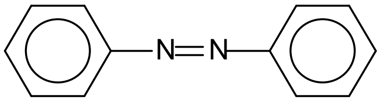
This document consists of **13** printed pages.

## Section A

Answer **all** the questions in this section.

**1** **(a)** Scientists in Germany have developed a liquid crystal elastomers (LCE)–based adhesive that uses UV light to switch and control its level of stickiness within seconds.

To control adhesion, the team used azobenzene, $C_{12}H_{10}N_2$ in the LCE as the light responsive molecule, which isomerises quickly from one state to another and changes size under UV light. This effect flexes the material enough to cause the microstructures to peel away from a surface and unstick, akin to how a gecko loses adhesion by moving its feet. When the light is removed, the material quickly recovers to its flat, sticky state.



azobenzene

**(i)** State the type of hybridisation that is present on the nitrogen atom and draw the shape of the hybrid orbitals around one of the nitrogen atom in azobenzene. Suggest the bond angle around this nitrogen atom. [3]

**(ii)** State the type of isomerism that is present in azobenzene and draw the two structures that illustrate the isomerism. [2]

**(iii)** Suggest, with a reason, the isomer that is responsible for the peeling of microstructures away from a surface. [1]

**(b)** *Use of the Data Booklet is relevant to this question.*

Compound **U,** $C_{17}H_{17}N_3O_3$ is not soluble in NaOH(aq) and HC*l*(aq). It gives 2 products when heated under reflux with acidified potassium dichromate. Orange potassium dichromate turns green during the process. Upon careful neutralisation of the resultant mixture, compounds **V** and **W** are formed.

**V,** $C_{13}H_{10}N_2O_2$ is a derivative of azobenzene, and has a proton ($^1$H) chemical shift value of 13.0 ppm.

**W** gives an orange precipitate when treated with 2,4–dinitrophenylhydrazine but it has no reaction with Tollens' reagent. It gives a yellow precipitate and compound **X** when warmed with aqueous mixture of sodium hydroxide and iodine. Effervescence was observed when **W** was treated with aqueous sodium hydrogencarbonate.

When **W** is heated with excess $CH_3Cl$, it gives compound **Y,** $C_7H_{14}NO_3Cl$, as the major product.

Suggest structures for **U** – **Y** and explain the reactions described above.
(The –N=N– structure remains unaltered in this question) [9]

[Total: 15]

**2** Silver forms a series of halides of general formula Ag*X*. The chloride, bromide and iodide of silver are sparingly soluble in water at room temperature.

Data about the solubilities in water and the solubility products of the chloride, bromide and iodide of silver at 298 K are given below.

| Salt | Solubility / mol $dm^{-3}$ | Solubility product / $mol^2\ dm^{-6}$ |
|---|---|---|
| $AgCl$ | $1.4 \times 10^{-5}$ | $2.0 \times 10^{-10}$ |
| $AgBr$ | $7.1 \times 10^{-7}$ | $5.0 \times 10^{-13}$ |
| $AgI$ | $8.9 \times 10^{-9}$ | to be calculated |

In this question, give **each** of your numerical answers to **one** decimal place.

**(a)** **(i)** Write an expression for the solubility product, $K_{sp}$ of silver iodide. [1]

**(ii)** From the data above, calculate a value for $K_{sp}$ of silver iodide. [1]

**(iii)** To a 2.0 $dm^3$ of saturated solution of $AgI$, 0.025 g of $AgNO_3(s)$ was added. Calculate the mass of precipitate formed. [2]

**(b)** When a precipitate is formed, $\Delta G^{\ominus}_{ppt}$ is given by the following expression.

$$\Delta G^{\ominus}_{ppt} = 2.303\ RT \log_{10} K_{sp}$$

**(i)** Use the data above to calculate $\Delta G^{\ominus}_{ppt}$, for silver chloride. [1]

**(ii)** When silver nitrate is added to solution containing chloride ions, a white precipitate is observed. The white precipitate dissolves when aqueous ammonia was added. Upon adding of aqueous sodium bromide to the resultant mixture, a cream precipitate is obtained.

With the aid of suitable equations, explain the chemistry that is occurring during these reactions. [3]

**(c)** For silver fluoride, $AgF$, $K_{sp}$ = 1.006 $mol^2\ dm^{-6}$ at 298 K.
Use the expression given in **(b)** to determine whether silver fluoride is soluble in water at 298 K.

Explain your answer. [2]

[Total: 10]

**3** **(a)** Great research efforts have been put into the production of sustainable fuels, such as hydrogen and synthetic hydrocarbons. One of particular interest is a solar–driven thermochemical process which utilises a metal oxide catalyst to produce carbon monoxide from carbon dioxide.

In the first stage, the metal oxide ($MO_2$) is reduced to release oxygen gas. In the second stage, carbon dioxide then oxidises the reduced metal oxide to generate carbon monoxide.

The 2–stage process is as shown below.

Stage 1: $MO_2(s) \rightleftharpoons MO_{2-x}(s) + \frac{x}{2}O_2(g)$ $\Delta H_1 > 0$

Stage 2: $MO_{2-x}(s) + xCO_2(g) \rightleftharpoons MO_2(s) + xCO(g)$

**(i)** Stage 1 is carried out at a high temperature of 1500 °C.

State *Le Chatelier's Principle* and use it to explain why such a high temperature is necessary. [2]

**(ii)** Give the equation, with state symbols, for the overall equilibrium reaction. [1]

**(iii)** The $\Delta H^{\ominus}$ of the reaction in **(a)(ii)** is +283 kJ mol$^{-1}$ when $x = 1$.

Using relevant data from the *Data Booklet*, calculate the carbon–oxygen bond energy in carbon monoxide. [2]

**(iv)** Given that the standard Gibbs free energy change, $\Delta G^{\ominus}$, of the reaction in **(a)(ii)** is +237 kJ mol$^{-1}$ at 298 K, calculate the $\Delta S^{\ominus}$ for this reaction and comment on its sign with respect to the reaction. [2]

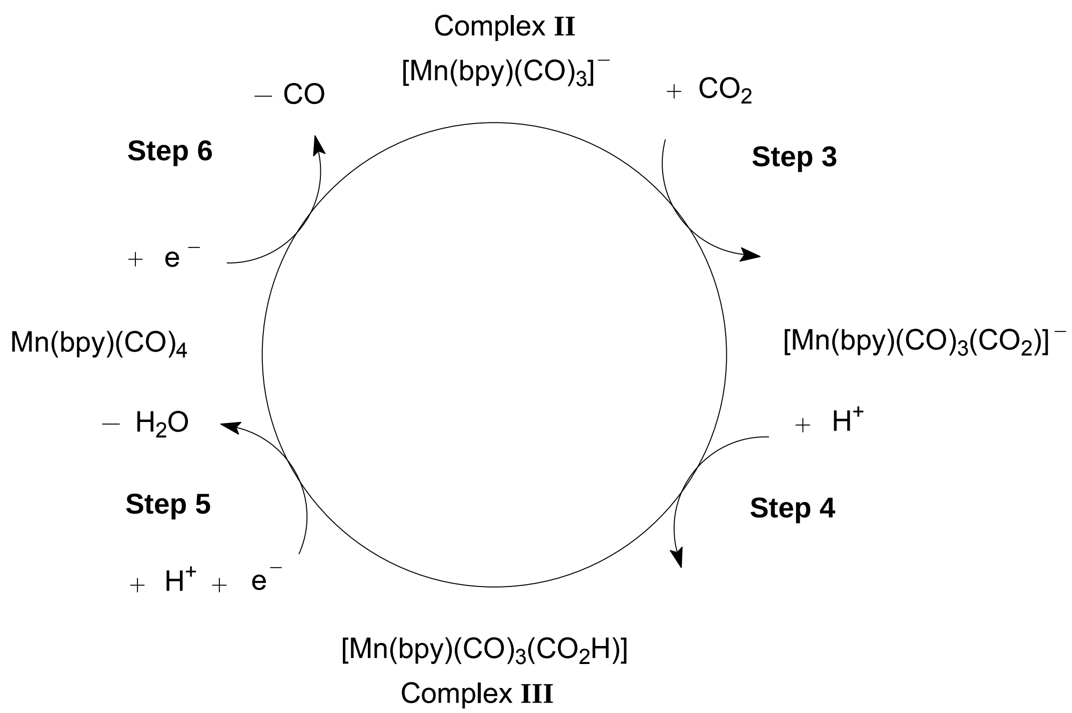
**(b)** The conversion of carbon dioxide to carbon monoxide is also important in helping to slow down global warming. Another method that was developed by scientists involves the use of metal complexes as catalysts to capture carbon dioxide and reduce it to carbon monoxide using visible light.

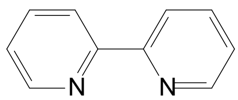
The starting reagent for the method involves the conversion of Complex **I**, $Mn(bpy)(CO)_3(Br)$ to the catalyst for the process, Complex **II**, $[Mn(bpy)(CO)_3]^-$.

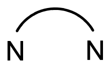
**Step 1**: $Mn(bpy)(CO)_3(Br) + e^- \longrightarrow [Mn(bpy)(CO)_3(Br)]^-$

**Step 2**: $[Mn(bpy)(CO)_3(Br)]^- + e^- \longrightarrow [Mn(bpy)(CO)_3]^- + Br^-$

Complex **II** then undergoes a catalytic reduction process with carbon dioxide as seen below.




where bpy = 


**(i)** Using  to represent the bpy ligand, draw the structure of Complex **I** and state its shape. [2]

**(ii)** State which of the six steps involve a **decrease** in the number of ligands bonded to the manganese ion. [1]

**(iii)** Suggest the type of reaction which takes place in steps 1 and 4. [1]

**(iv)** A solution containing Complex **III** is orange in colour.

Explain how such a colour could arise. [3]

**(c)** The CarbFix project in Iceland is also targeted at alleviating the global problem of increasing industrial carbon dioxide ($CO_2$) emissions and climate change. All waters in contact with the atmosphere absorb $CO_2$, but this method speeds up the process by injecting waters containing $CO_2$ into basaltic rocks which are rich in divalent cations such as calcium and magnesium. The method is called "stoning" as the $CO_2$ is captured and stored in the rocks as insoluble carbonates.

Water containing $CO_2$ reacts with the divalent cations present in basaltic rocks according to the following equilibrium.

$$M^{2+} + CO_2 + H_2O \rightleftharpoons MCO_3 + 2H^+ \quad \text{———— (1)}$$

**(i)** Explain why the precipitation of the metal carbonate takes place at high pH. [1]

**(ii)** Water containing $CO_2$ also reacts with Group 2 carbonates to form the corresponding soluble hydrogencarbonates in the following equilibrium.

$$CO_2(g) + H_2O(l) + MCO_3(s) \rightleftharpoons M(HCO_3)_2(aq) \quad \text{———— (2)}$$

These soluble compounds are then washed away by rainwater that percolates through rocks on hills and mountains and emerges as springs in the hillside. These are known as mineral waters and they often contain chlorides and hydrogencarbonates of Group 2 metals.

One such mineral water has the following composition.

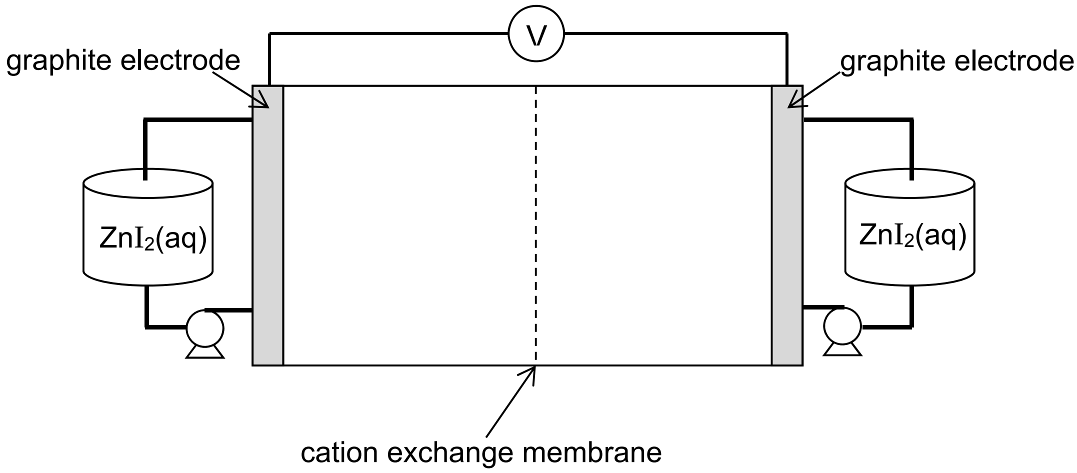
| **ion** | **concentration / g dm$^{-3}$** |
|---|---|
| $Ca^{2+}$ | 0.0120 |
| $Mg^{2+}$ | 0.0073 |
| $Cl^-$ | 0.0284 |
| $HCO_3^-$ | 0.0244 |

Calculate the concentrations of these ions in mol dm$^{-3}$ and hence suggest the formulae of the salts that could exist in the solution, and their relative amounts. [3]

**(iii)** Suggest the composition of the rock through which the rainwater had percolated and use equation (2) to explain how the mineral water had been formed. [2]

[Total: 20]

**4** Redox flow batteries are rechargeable batteries that have received growing attention as a cost–effective energy storage solution. Compounds with iodine can be used to generate electrical energy in redox flow batteries. One such battery is the zinc–iodide battery shown below.




In the zinc–iodide flow cell, graphite is used as the electrodes and the two half–cells are separated by a cation exchange membrane which allows cations to flow through. The electrolyte, $ZnI_2$(aq), is stored in two external tanks. When required, the electrolyte is pumped into both half–cells.

During charging, metallic zinc is deposited on the negative electrode and iodine is formed at the positive electrode. When a light bulb is connected across the two electrodes, the bulb lights up as the battery discharges.

**(a)** **(i)** By reference to the *Data Booklet*, choose two half–equations to construct the balanced equation for the reaction that occurs during the discharging process. [2]

**(ii)** Calculate the value for $E^{\ominus}_{cell}$ for this zinc–iodide battery. [1]

**(iii)** Hence calculate the $\Delta G^{\ominus}$ for this cell. [1]

**(iv)** Suggest why there is a need for a cation exchange membrane in the battery. [1]

**(b)** The iodine formed during charging can complex with the iodide ions in the electrolyte to form triiodide ions, $I_3^-$. This reduces the efficiency of the battery, as formation of triiodide ions reduces the amount of $I^-$ available for redox reaction.

Recent research has shown that addition of bromide, $Br^-$, to the zinc–iodide battery frees up the iodide ions, as bromide can form a similar complex with iodine.

Draw a dot and cross diagram of the complex formed between iodine and bromide ion and state its shape. [2]

**(c)** Compounds with a halogen and an alcohol group are known as halohydrins. Halohydrins are useful intermediates in organic synthesis as they contain two reactive functional groups.

**(i)** Halohydrins can be synthesised from the reaction of alkenes with aqueous bromine.

Describe the mechanism of the reaction between 1–methylcyclohexene and aqueous bromine to form bromohydrin. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons. [4]

**(ii)** The bromohydrin formed in **(c)(i)** exists as a mixture of stereoisomers. Draw the structure of each stereoisomers of the bromohydrin formed. [2]

**(iii)** Suggest one simple test–tube reaction, other than aqueous bromine, to distinguish 1–methylcyclohexene and the bromohydrin formed in **(c)(i)**, stating clearly the reagents and conditions required and the observations you would see. [2]

[Total: 15]

## Section B

Answer **one** question from this section.

**5** The oxygen family, also called the chalcogens, consists of the elements found in Group 16 of the Periodic Table and is considered among the main group elements. It consists of the elements oxygen, sulfur, selenium, tellurium and polonium.

**(a)** State and explain the trend in the first ionisation energy of the Group 16 elements down the group. [2]

**(b)** **(i)** State two assumptions of the kinetic theory of gases. [2]

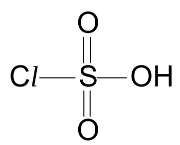
**(ii)** Sketch the expected variations of *p*V with *p* for a given amount of an ideal gas at constant temperature. [1]

In an experiment conducted at constant temperature, the volumes of separate samples containing equal amounts of gases **D** and **E** were measured at different pressures and the results are tabulated below. The two unknown gases **D** and **E** could be hydrogen or oxygen.

| **Experiment No.** | **Gas D** | | **Gas E** | |
|---|---|---|---|---|
| | ***p* / Pa** | **V / m³** | ***p* / Pa** | **V / m³** |
| 1 | $3.5 \times 10^{5}$ | $6.50 \times 10^{-3}$ | $4.0 \times 10^{5}$ | $5.80 \times 10^{-3}$ |
| 2 | $7.0 \times 10^{5}$ | $3.14 \times 10^{-3}$ | $8.0 \times 10^{5}$ | $2.85 \times 10^{-3}$ |
| 3 | $14.0 \times 10^{5}$ | $1.50 \times 10^{-3}$ | $15.0 \times 10^{5}$ | $1.46 \times 10^{-3}$ |
| 4 | $21.0 \times 10^{5}$ | $9.30 \times 10^{-4}$ | $20.0 \times 10^{5}$ | $1.07 \times 10^{-3}$ |

**(iii)** Compute the *p*V values for each of the above experiments and use them to identify gases **D** and **E**. Explain your reasoning. [3]

**(c)** The diagram below shows the structure of chlorosulfuric acid, $HSO_3Cl$, which is a Lewis base.




chlorosulfuric acid

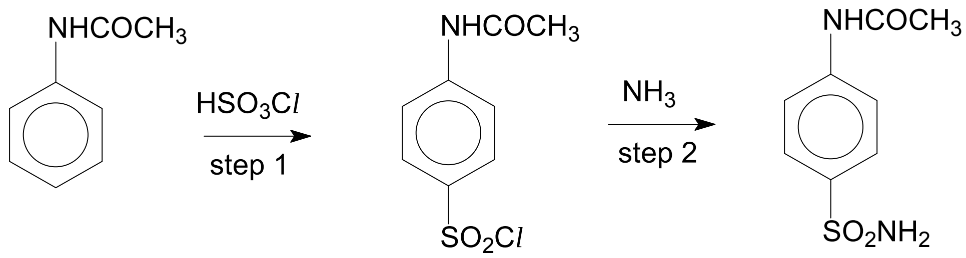
**(i)** What do you understand by the term *Lewis base*? [1]

**(ii)** Chlorosulfuric acid is made by reacting sulfuric acid with phosphorus pentachloride, $PCl_5$. Acidic white fumes are evolved during the reaction.

Write a balanced equation for the reaction between sulfuric acid and phosphorus pentachloride. [1]

**(iii)** By comparing the structure of chlorosulfuric acid with that of sulfuric acid, explain why sulfuric acid is expected to be a stronger acid than chlorosulfuric acid. [1]

**(d)** The following reaction scheme shows the synthesis of **C**, a precursor to the sulfonamide antibacterial drugs.




**A** **B** **C**

**(i)** Explain why compound **A** is neutral. [1]

**(ii)** $C_6H_5CONHCH_3$ is an isomer of **A**. Describe a simple chemical test to distinguish between the two isomers, stating clearly how each compound behaves in the test. [2]

Step 1 is catalysed by concentrated sulfuric acid.

**(iii)** Write an equation to illustrate the reaction between concentrated sulfuric acid and chlorosulfuric acid, $HSO_3Cl$. [1]

**(iv)** Hence, name and describe the mechanism for the reaction involved in the preparation of **B**, including curly arrows to show the movement of electrons, and all charges. [3]

Step 2 is carried out under anhydrous condition.

**(v)** What type of reaction is step 2? [1]

**(vi)** Suggest why step 2 has to be carried out under anhydrous condition. [1]

[Total: 20]

**6** **(a)** In acidic solution, bromate(V) ions, $BrO_3^-$, slowly oxidise bromide ions to bromine. The progress of the reaction may be followed by adding a fixed amount of phenol together with some methyl red indicator.

The bromine produced during the reaction reacts very rapidly with phenol. When all the phenol is consumed, any further bromine bleaches the indicator immediately. The initial rate of formation of $Br_2$ is indicated by the time for the bromine to bleach the indicator.

The total volume of the reaction mixture is the same in all four experiments and the following kinetic data are obtained at 25 °C.

| **Experiment** | **[$BrO_3^-$] / mol dm$^{-3}$** | **[$Br^-$] / mol dm$^{-3}$** | **pH** | **Initial rate of formation of $Br_2$ / mol dm$^{-3}$ s$^{-1}$** |
|---|---|---|---|---|
| 1 | 0.10 | 0.10 | 1.00 | $8 \times 10^{-2}$ |
| 2 | 0.10 | 0.05 | 1.00 | $4 \times 10^{-2}$ |
| 3 | 0.05 | 0.05 | 1.00 | $2 \times 10^{-2}$ |
| 4 | 0.05 | 0.10 | 1.30 | $1 \times 10^{-2}$ |

**(i)** Write a balanced equation for the reaction between bromide ion and bromate(V) ion. [1]

**(ii)** Determine the order of the reaction with respect to each of the following reactants

- $BrO_3^-$
- $Br^-$
- $H^+$. [2]

**(iii)** Write the rate equation for this reaction. [1]

**(iv)** Calculate the rate constant of the reaction at this temperature, stating its units. [1]

**(b)** Describe, and explain in molecular terms, how the rate of reaction is affected by an increase in temperature. You should include a reference to the Boltzmann distribution in your answer. [3]

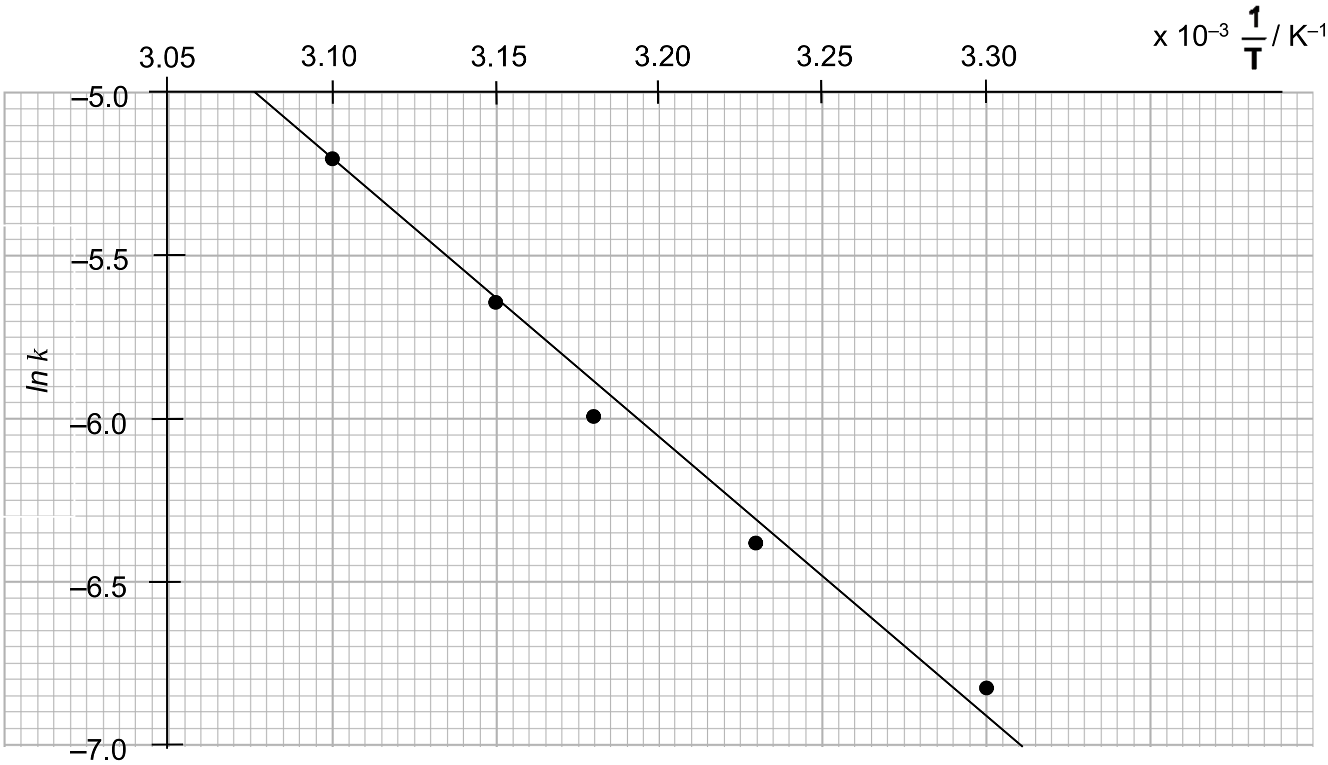
**(c)** The effect of temperature on the rate constant, $k$, can be expressed by the following equation.

$$\ln k = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right)$$

A is a constant;
$E_a$ is the activation energy;
R is the molar gas constant (8.31 J $K^{-1}$ $mol^{-1}$) and
T is the absolute temperature.

In a separate experiment to investigate the effect of temperature on the rate constant of the reaction between bromide ion and bromate(V) ion, the graph below was obtained.

Graph of $\ln k$ against $\frac{1}{T}$




**(i)** Calculate the activation energy, $E_a$, of the reaction. [1]

**(ii)** Suggest why the activation energy of the reaction is high. [1]

**(iii)** Make a sketch of the graph shown above onto your writing paper. On the same axes, sketch the graph of $\ln k$ versus $\frac{1}{T}$ when the reaction proceeds in the presence of a catalyst. Label your graphs clearly. Explain your answer. [1]

**(d)** This question is about compound **K**, $C_6H_7ON$, which is formed when phenylhydroxylamine, $C_6H_5NHOH$, is warmed with dilute sulphuric acid.

- Compound **K** is not very soluble in water, but dissolves in HC*l*(aq).
- It also dissolves in NaOH(aq), but not in $Na_2CO_3$(aq).
- On reaction with 1 mol of ethanoyl chloride, $CH_3COCl$, **K** forms compound **L**, $C_8H_9O_2N$.

On reaction with $Br_2$ (aq), **L** produces compound **M**, $C_8H_7O_2NBr_2$. When **K** is reacted with 2 mol of ethanoyl chloride, it produces compound **N**, $C_{10}H_{11}O_3N$, which is not soluble in either HC*l*(aq) or NaOH(aq).

Compound **K** can be synthesised by treating phenol with dilute nitric acid, followed by reaction with tin metal and hydrochloric acid.

**(i)** Deduce the structures of compounds **K**, **L**, **M** and **N**. Explain the chemistry of the reactions described, writing equations where appropriate. [There is no need to comment on the chemistry of the formation of **K** from phenylhydroxylamine.] [7]

**(ii)** The reaction of **K** to form **L** produces a minor product **P**, which is a constitutional isomer of **L**. Suggest the structure of **P** and explain this observation. [2]

[Total: 20]